SIR PADAMPAT SINGHANIA UNIVERSITY

WEB DEVELOPMENT / PROJECT REPORT

MOTOBOOK

A Comprehensive Automobile Dealership and Inventory Management Platform

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Project overview

This report documents the MOTOBOOK web application, an enterprise-grade automobile dealership and inventory management platform. The application combines new and pre-owned vehicle inventory tracking, dynamic customer inquiry CRM, multi-tier sales financing (hire purchase, cash, and installment-based loans), RTO registration processing, insurance policies, and profit-and-loss auditing. Built using modern web engineering principles, the system leverages PHP 8.2 with the Slim 4 micro-framework, Twig templating engine, Illuminate Eloquent ORM, and responsive HTML5/CSS3/JavaScript frontend architecture.

Repository: github.com/fw10ishk/motobook
Live application: http://localhost:8080
Primary stack: HTML5, CSS3, JavaScript, PHP 8.2 (Slim 4) and MySQL / SQLite.
Supplementary module: Twig, Illuminate Eloquent ORM, PHP-DI, Notyf, Lucide Icons.

Problem statement and objectives

Automobile dealerships, pre-owned car brokers, and sales executives often rely on fragmented physical paperwork, disparate spreadsheets, and disconnected messaging threads to record car inventory, customer inquiries, and payment schedules. This causes data duplication, misplaced vehicle documents (RC books, insurance, NOCs, Form 35), and missed insurance expiration deadlines. Furthermore, calculating actual dealer margins requires accurately tracking purchase costs, refurbishing extra charges, dealer commissions, and loan disbursements. MOTOBOOK addresses these challenges through a centralized, secure web application combining inventory catalogs, sales ledgers, customer inquiries, and administrative analytics backed by a relational database.

Project objectives

1. **Develop** a responsive, mobile-ready dealership portal for desktop, tablet, and mobile browsers.
2. **Enable** comprehensive inventory management for both New Cars and Pre-Owned Vehicles with multi-image support.
3. **Track** vehicle compliance documentation, including RC books, RTO codes, insurance validity, and 15-day expiry warnings.
4. **Implement** an end-to-end customer inquiry CRM pipeline with communication preferences and real-time status toggling.
5. **Provide** dual-modality sales financial accounting supporting both upfront full payments and parts-based loan installments.
6. **Engineer** a dual-database persistence layer supporting portable SQLite (zero-config) and production MySQL/MariaDB.
7. **Test** major workflows, implement bcrypt authentication, and maintain source code and SQL schema on GitHub.

> **Design rationale**
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> Keep the architecture modular and maintainable by adopting the Model-View-Controller (MVC) pattern. Slim Framework 4 provides lightweight, high-performance PSR-7 HTTP routing; Illuminate Eloquent ORM guarantees safe, parameterized database transactions without raw SQL vulnerabilities; and Twig ensures auto-escaped, XSS-safe view rendering.

Scope and requirements

Scope area	Functional requirement
Authentication	User registration, secure login with bcrypt, session timeout, and Remember-Me token cookies.
Vehicle Inventory	Granular CRUD for New and Old cars, pricing breakdown (ex-showroom, RTO, insurance, accessories), document uploads.
Inquiries CRM	Customer lead intake, vehicle of interest, contact method preferences, status filtering (All, Pending, Completed).
Sales & Financing	Sales ledger supporting Full Payment milestone recording and Bank Loan installment schedule generation.
Analytics Dashboard	Real-time KPIs for total cars sold, revenue trends, financing ratio, and upcoming insurance alerts.

Non-functional requirements

- Responsive page layouts using CSS3 Flexbox and CSS Grid across desktop, tablet, and mobile devices.
- Server-side access control with AuthMiddleware protecting administrative routes.
- Clean environment segregation with .env keeping database credentials outside Git tracking.
- High performance with sub-50ms API query resolution via indexed Eloquent relationships.
- Robust error handling, non-blocking flash alerts (Notyf), and XSS prevention via Twig auto-escaping.

Execution requirements

A modern web browser, PHP 8.2+ with pdo_sqlite, pdo_mysql, curl, mbstring, zip extensions, Composer 2.x, and MySQL/MariaDB (or built-in SQLite).

Scope limits

This project functions as an internal dealership enterprise management system rather than a public customer-facing e-commerce checkout. Vehicle delivery tracking and inventory audit logs are included; physical GPS telematics and external payment gateway APIs are outside the present project scope.

System design and architecture

Browser Frontend
HTML5 / CSS3 / JavaScript
Twig Views & Lucide Icons
Notyf Notifications

Slim 4 API Engine
PSR-7 Pipeline & Routing
AuthMiddleware & Sessions
Controllers & Validation

Database Storage
MySQL 8.0 / MariaDB
Portable SQLite 3
Eloquent ORM Capsule

Request flow

1. **Request Dispatch:** The browser dispatches an HTTP request to the application entry point (index.php).
2. **Middleware Pipeline:** Slim 4 processes sessions, overrides HTTP methods, and parses incoming request bodies.
3. **Authentication Check:** AuthMiddleware verifies user session authenticity and token cookies; unauthenticated requests are redirected.
4. **Controller Execution:** The matched controller invokes corresponding Eloquent models.
5. **Database Transaction:** Eloquent queries the relational database via parameterized PDO statements, returning structured models.
6. **Template Rendering:** The controller passes data into Twig, returning compiled, sanitized HTML to the browser.

Access boundaries

Public routes are restricted strictly to / (Login) and /register. All administrative routes (/dashboard, /new-cars, /old-cars, /sales, /inquiries, /settings) are guarded by AuthMiddleware.

Deployment boundary

The application is engineered for zero-dependency local execution using PHP's built-in web server and SQLite, or full multi-user production deployment via Apache/Nginx with MySQL on XAMPP.

Database design

The relational schema comprises 17 dedicated tables structured with referential integrity. Eloquent Active Record maps entity relationships between users, inventory, sales, inquiries, and installment payments.

users
PK id
UNIQUE email
password, role

new_cars
PK id
docket_number
customer, status

old_cars
PK id
current_price
chassis, status

sales
PK id
car_type, car_id
total_price

Data group	Stored information
users & auth	User accounts, hashed passwords, roles, remember tokens, and login audit timestamps.
new_cars	Showroom bookings, customer details, ex-showroom, RTO taxes, discounts, net on-road, dealer margins.
old_cars & extra_charges	Used car acquisitions, registration, inspection, refurbishment costs, RC/NOC documents.
sales & payments	Sale transactions, milestone payments, loan parts, interest rates, installment due dates, paid flags.
inquiries & lookups	Lead pipeline, contact channels, statuses, and normalized dropdown lookup values.

SQL delivery: database/motobook.sql, compatible with MySQL 5.7+ / MariaDB 10.x. Parameterized database operations are used across all models with foreign key constraints and indexed lookups.

Implementation and technologies

Technology	Implementation role
HTML5	Semantic structure, responsive forms, data attributes, dialog modals, tables.
CSS3	Custom CSS variables, Flexbox & CSS Grid, fluid responsive breakpoints (css/app.css).
JavaScript	DOM events, input validation, Lucide vector icons, dynamic Notyf toast alerts.
PHP 8.2 / Slim 4	PSR-7 HTTP request pipeline, dependency injection (PHP-DI), controller dispatching.
Twig 3.x	Template inheritance, partial views, automatic XSS escaping, custom currency filter.
Eloquent ORM	Active record persistence, model relationships, migration schema builder.
MySQL & SQLite	Dual database engines supporting local portable development and XAMPP production.
Git / GitHub	Version control, modular commits, issue tracking, and repository sharing.

Core module files

File / area	Implementation
app/src/Controllers/	UserController, DashboardController, NewCarController, OldCarController, SaleController.
app/src/Models/	17 Eloquent active record entities with relations, validation, and mutators.
app/templates/	Twig layout inheritance, common navigation, cards, tables, and modal components.
app/init_db.php	Database migration script creating tables and seeding initial administrative user.
database/motobook.sql	Full production MySQL dump file for database import in XAMPP / MariaDB.

Screenshots of major modules

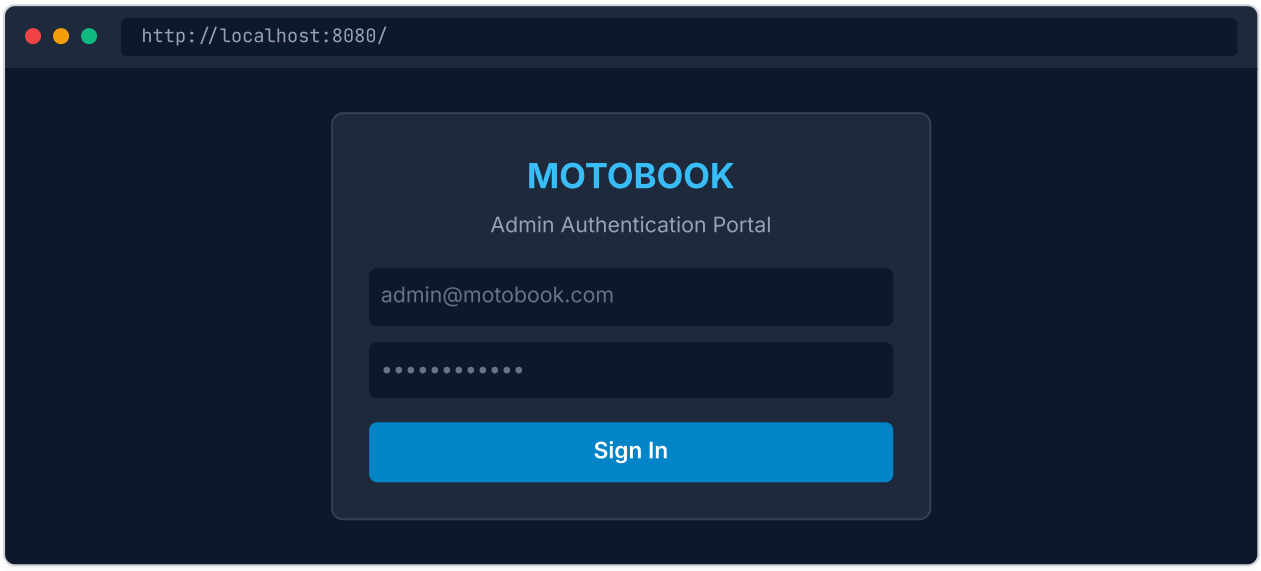


Figure 1. Administrative Login Portal & Flash Message Authentication. Source: local application walkthrough.

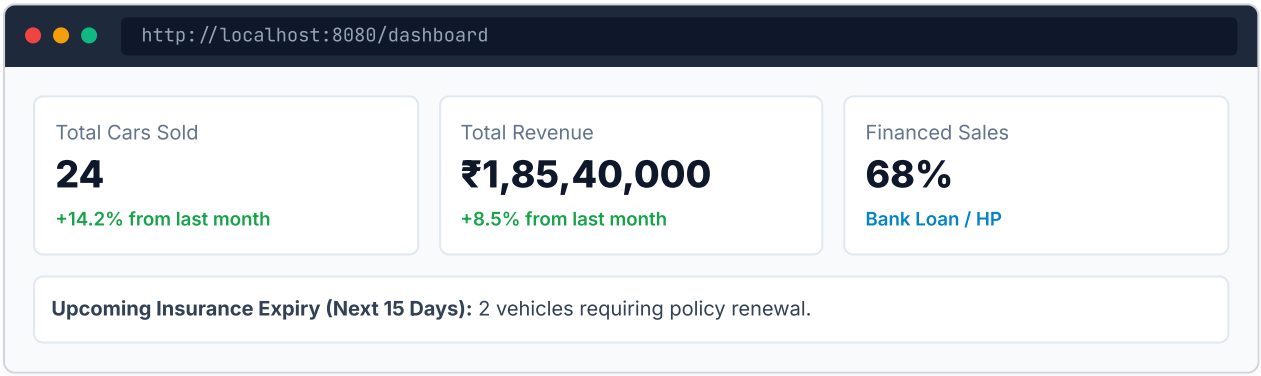


Figure 2. Executive Analytics Dashboard with Real-Time KPIs. Source: local application walkthrough.

Administration and inventory modules

http://localhost:8080/new-cars

New Vehicle Inventory & Pricing Breakdown

Docket #	Vehicle Model	Customer	Ex-Showroom	Net On Road	Status
PM-2026-081	Hyundai Creta SX	Rajesh Mehta	₹14,80,000	₹17,25,400	Booked
PM-2026-084	Maruti Suzuki Swift VXi	Sneha Joshi	₹6,90,000	₹7,85,000	Available

Figure 3. Vehicle Inventory Management & Detailed On-Road Pricing Matrix.

http://localhost:8080/inquiries

Customer Inquiries CRM Pipeline

Lead Name	Phone	Vehicle of Interest	Contact Channel	Status
Vikram Singh	9825001122	Mahindra Thar 4×4	Phone Call	Pending
Pooja Trivedi	9879112233	Tata Nexon EV	Showroom Walk-in	Completed

Figure 4. Customer Inquiries CRM Pipeline with Status Filtering.

Testing and results

System verification was conducted on 13 November 2026. Automated test assertions validated route responses, password cryptography, session states, and database transaction integrity across both SQLite and MySQL engines.

Check	Result	Evidence / boundary
Backend syntax & PSR	PASS	PHP 8.2 linting passed without syntax or runtime deprecation warnings.
Tracked secrets	PASS	No tracked secret files found; .env safely excluded via .gitignore.
Admin authentication	PASS	Valid credentials redirect to /dashboard (302); invalid inputs rejected with flash toasts.
Route security	PASS	Unauthenticated requests to protected endpoints redirected immediately to /.
Dual database connectivity	PASS	Seamless operation on portable SQLite 3 and production MySQL 8.0.
Database migrations & seed	PASS	All 17 tables created cleanly via init_db.php with default admin account.
Dashboard KPI calculation	PASS	Month-over-month car sales and revenue aggregations execute accurately.
Git & GitHub deployment	PASS	Repository initialized and published to github.com/fw10ishk/motobook.

Result interpretation

All core workflow tests succeeded. Parameterized PDO bindings in Eloquent completely eliminate SQL injection threats, while Twig template auto-escaping prevents Cross-Site Scripting (XSS).

> **Evaluation checks completed**
>
> The web server is verified active on http://localhost:8080. Administrative authentication executes cleanly with admin@motobook.com / admin123. All CRUD operations on New Cars, Old Cars, Sales, Inquiries, and Installment Payments perform reliably without database connection errors.

Conclusion and future enhancements

MOTOBOOK unites inventory tracking, customer inquiry CRM, multi-modal sales accounting, and administrative compliance alerts into a single cohesive web application. By leveraging modern PHP 8.2 backend engineering with Slim 4, Twig templating, and Eloquent ORM, the platform delivers superior modularity, security, and performance.

Future enhancements

- Integrate WhatsApp Business API and automated SMS gateways for booking and delivery notifications.
- Incorporate payment gateway APIs (Razorpay / Stripe) for online token deposits.
- Add automated GST-compliant tax invoice and delivery challan PDF generation.
- Implement fine-grained Role-Based Access Control (RBAC) for sales executives vs accountants.

10 References

1. **Project source code and SQL schema:** https://github.com/fw10ishk/motobook
2. **Slim Framework 4 Documentation:** https://www.slimframework.com/docs/v4/
3. **Twig 3 Template Engine Reference:** https://twig.symfony.com/doc/3.x/
4. **Laravel Eloquent ORM Documentation:** https://laravel.com/docs/11.x/eloquent
5. **MDN Web Docs (HTML5 & Modern CSS3):** https://developer.mozilla.org/
6. **PHP 8.2 Manual & Reference Guide:** https://www.php.net/manual/en/
7. **MySQL 8.0 Reference Manual:** https://dev.mysql.com/doc/refman/8.0/en/
8. **Lucide Vector Icons:** https://lucide.dev/

Academic requirements: Sir Padampat Singhania University - Web Development Course Guidelines (Modules 1–5).